

Rename `system_context_replaceability` namespace

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§ Abstract

This paper proposes to rename `system_context_replaceability` namespace.

§ 1 Motivation

There was one attempt to rename `system_context_replaceability`. Unfortunately, it didn't go well because likely the proposed alternatives for the names were not received well. [\[P3804R1\]](#) proposed several options favoring `replacement` and `replacement_functions`. As people pointed out, those names are probably also not great because they don't give any sense of what developers try to replace. For example `std::execution::replacement::receiver_proxy` doesn't help to understand what we're actually replacing. The motivation to recommend either `replacement` or `replacement_functions` came because `system_context_replaceability` felt too long for a namespace name. However, people don't seem to care about the length of the name. The poll below taken in Kona meeting (2025) shows that:

POLL: Change namespace `system_context_replacability` to `replacement` as proposed in [P3804R0](#) Iterating on `parallel_scheduler`.

SF	F	N	A	SA
2	1	6	5	1

What is also important to note is that the poll was only about changing the current status quo to replacement. We have not voted for other options in [P3804R1]. Nevertheless, in my opinion we have to rename `system_context_replaceability` to something else because what we should care about is the consistency of C++ standard (whenever possible). Adding `system_context_replaceability` namespace name to C++26 will cause a lot of confusion because `system_context` does not even exist in the current working draft. Thus, users would not understand what they are actually replacing. What we have in the current working draft is actually named `parallel_scheduler` and the replaceability feature was developed specifically for it. I understand the argument that suggested replacement option perhaps does not give us what we want but it doesn't eliminate the fact that we should rename `system_context_replaceability` to avoid even bigger confusion because `std::execution::system_context_replaceability::receiver_proxy` with no `system_context` presence anywhere in the C++ working draft is no better.

Based on the above either of `parallel_scheduler_replaceability` or `parallel_scheduler_replacement` name should be good enough because people seem to not be bothered by the name length and also because `parallel_scheduler` is exactly the name we have for the scheduler itself in the current working draft. Leaving the status quo would be a disaster, in my opinion.

§ 2 Proposal

Option 1: Change `std::execution::system_context_replaceability` namespace name to `std::execution::parallel_scheduler_replaceability`.

Option 2: Change `std::execution::system_context_replaceability` namespace name to `std::execution::parallel_scheduler_replacement`.

Modify the formal wording in accordance with the winning option (if any wins compared to the status quo).

I like the second option slightly better because *replaceability* word means the feature to me and *replacement* sounds more like a set of the actual API. But I can live with any of those.

§ 3 References

[P3804R1] Lucian Radu Teodorescu, Ruslan Arutyunyan. 2025-12-15. Iterating on `parallel_scheduler`.

<https://wg21.link/p3804r1>